Gearbox shaft axial end float

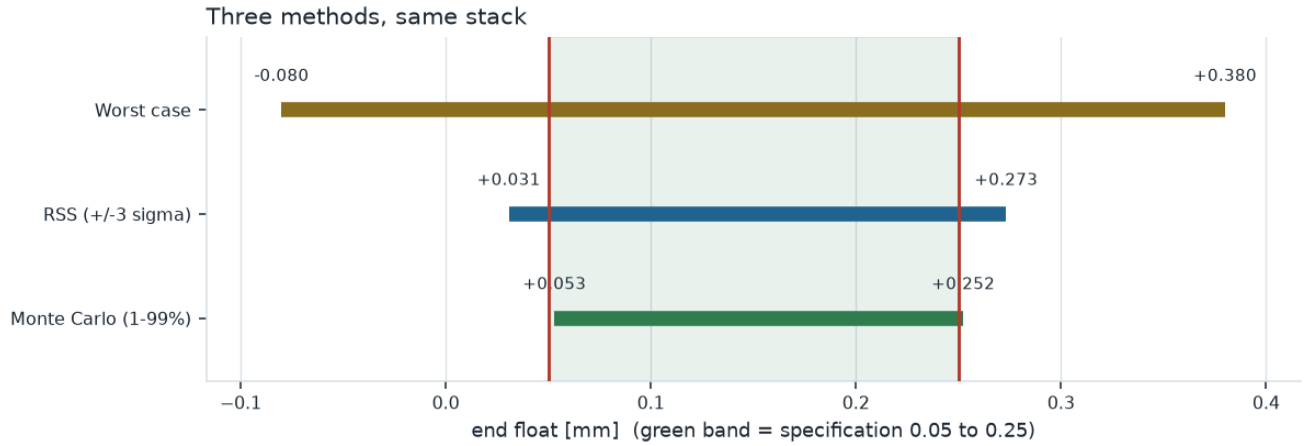
Axial stack setting the end float of a shaft carried on two deep-groove ball bearings, located by a spacer and a ground shim inside a housing bore. End float must stay positive (no axial preload on the bearings) and bounded (no excessive play). · closing dimension end float, specification 0.05 to 0.25 mm · 5 contributors · tolerance = 3s · generated 2026-11-03 10:31

Cpk 0.76, 19,860 ppm predicted scrap

Monte Carlo over 200,000 runs: mean 0.152 mm, sigma 0.0429 mm, yield 98.01%. Nominal gap 0.150 mm.

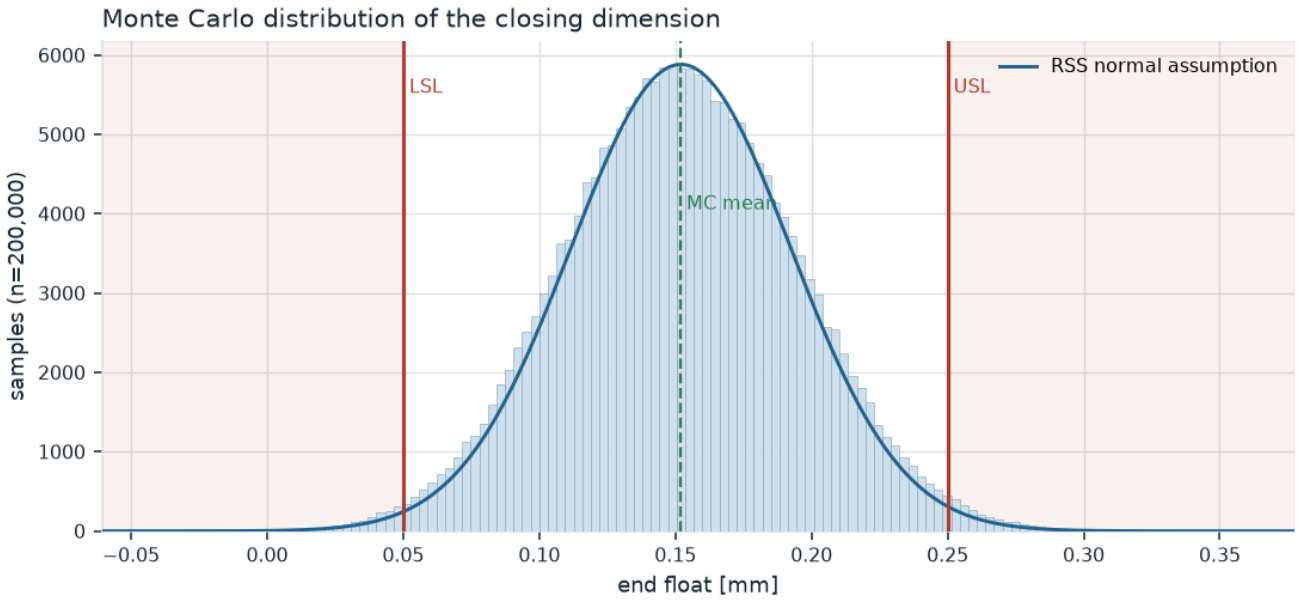
0.150	0.0429	98.01%	0.76
nominal gap	MC sigma	yield	Cpk (MC)

Worst case does not fit: the extremes run -0.080 to 0.380 mm against a 0.05 to 0.25 mm specification. That alone is not a reason to change the design — the probability of every part sitting at its limit together is negligible — but it does mean the stack has to be justified statistically rather than by inspection.



Three methods

Monte Carlo disagrees with RSS. RSS predicts 13,461 ppm, Monte Carlo 19,860 ppm — worse by 6,399 ppm. RSS assumes every contributor is normal and centred; the stack declares a skewed bearing width, a uniform shim and a mean shift on the bore, and that is the whole of the difference.

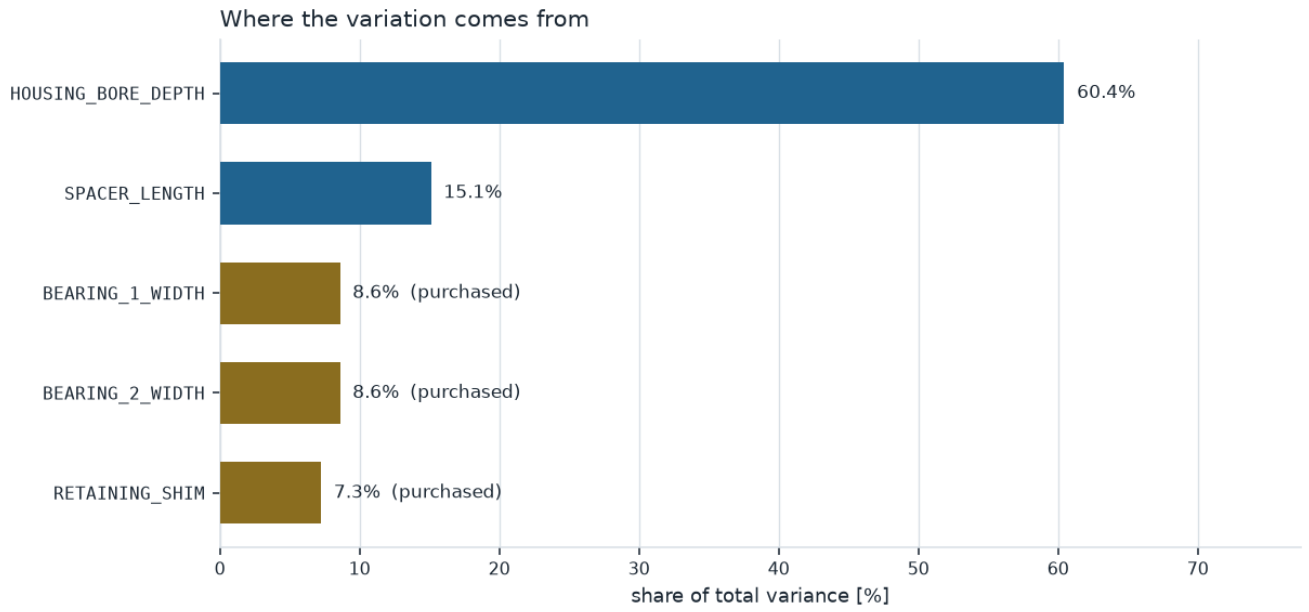


method	mean	half-band / 3s	low	high	predicted scrap	Cpk
worst case	0.1500	0.2300	-0.0800	0.3800	—	—
RSS	0.1520	0.1212	0.0308	0.2732	13,461 ppm	0.81
Monte Carlo	0.1519	0.1286	0.0530	0.2524	19,860 ppm	0.76

Rejects split 1,664 low (8,320 ppm) and 2,308 high (11,540 ppm). Which tail dominates decides the fix: a lopsided split is a nominal problem, an even split is a variation problem.

Contributors

HOUSING_BORE_DEPTH carries 60% of the variance. Tolerances add as squares, so effort spent anywhere else is close to wasted until this one moves: halving its tolerance removes about 45% of the total variance on its own.



name	nominal	tolerance	dir	distribution	s	variance share	source
HOUSING_BORE_DEPTH	60.000	-0.100 / +0.100	+1	normal, shift +0.3s	0.0333	60.4%	machined in-house
SPACER_LENGTH	24.950	-0.050 / +0.050	-1	normal	0.0167	15.1%	machined in-house
BEARING_1_WIDTH	15.000	-0.030 / +0.030	-1	triangular (mode +0.012)	0.0126	8.6%	purchased, DIN 620 normal class
BEARING_2_WIDTH	15.000	-0.030 / +0.030	-1	triangular (mode +0.012)	0.0126	8.6%	purchased, DIN 620 normal class
RETAINING_SHIM	4.900	-0.020 / +0.020	-1	uniform	0.0115	7.3%	purchased, ground shim stock

Tolerance allocation

Reaching Cpk 1.33 needs the in-house tolerances scaled by ×0.508 (total sigma budget 0.0246 mm).

contributor	current	required
HOUSING_BORE_DEPTH	-0.1000 / +0.1000	-0.0508 / +0.0508
SPACER_LENGTH	-0.0500 / +0.0500	-0.0254 / +0.0254

Held fixed (purchased): BEARING_1_WIDTH, BEARING_2_WIDTH, RETAINING_SHIM Worst case sums the extreme limits. RSS assumes each contributor is normal and centred. Monte Carlo samples the declared distributions with a fixed seed. All sensitivity coefficients are ±1: this is a linear 1D stack, so no geometric or kinematic effects are modelled. Form and orientation tolerances, temperature and assembly-induced deflection are outside the model.